



MHH772 / 90DaysOfDevOps



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90DaysOfDevOps / 2026 / day-15



LondheShubham153 Added day 15 tasks

929b190 · 7 months ago



This branch is 14 commits ahead of TrainWithShubham/90DaysOfDevOps:master



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README.md

Added day 15 tasks

7 months ago

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Day 15 – Networking Concepts: DNS, IP, Subnets & Ports

Task

Build on Day 14 by understanding the building blocks of networking every DevOps engineer must know.

You will:

- Understand how **DNS** resolves names to IPs
- Learn **IP addressing** (IPv4, public vs private)
- Break down **CIDR notation** and **subnetting** basics
- Know common **ports** and why they matter

This is concept-focused — research, understand, and document in your own words.

Expected Output

- A markdown file: `day-15-networking-concepts.md`

Challenge Tasks

Task 1: DNS – How Names Become IPs

1. Explain in 3–4 lines: what happens when you type `google.com` in a browser?
2. What are these record types? Write one line each:
 - `A` , `AAAA` , `CNAME` , `MX` , `NS`
3. Run: `dig google.com` — identify the A record and TTL from the output

Task 2: IP Addressing

1. What is an IPv4 address? How is it structured? (e.g., `192.168.1.10`)
2. Difference between **public** and **private** IPs — give one example of each
3. What are the private IP ranges?
 - `10.x.x.x` , `172.16.x.x – 172.31.x.x` , `192.168.x.x`
4. Run: `ip addr show` — identify which of your IPs are private

Task 3: CIDR & Subnetting

1. What does `/24` mean in `192.168.1.0/24` ?
2. How many usable hosts in a `/24` ? A `/16` ? A `/28` ?
3. Explain in your own words: why do we subnet?
4. Quick exercise — fill in:

CIDR	Subnet Mask	Total IPs	Usable Hosts
<code>/24</code>	?	?	?
<code>/16</code>	?	?	?
<code>/28</code>	?	?	?

Task 4: Ports – The Doors to Services

1. What is a port? Why do we need them?
2. Document these common ports:

Port	Service
22	?
80	?
443	?
53	?
3306	?
6379	?
27017	?

3. Run `ss -tulpn` — match at least 2 listening ports to their services

Task 5: Putting It Together

Answer in 2–3 lines each:

- You run `curl http://myapp.com:8080` — what networking concepts from today are involved?
- Your app can't reach a database at `10.0.1.50:3306` — what would you check first?

Documentation

Create `day-15-networking-concepts.md` with:

- Your answers to each task
- Command outputs from `dig` and `ss`
- The filled CIDR table
- What you learned (3 key points)

Submission

1. Add `day-15-networking-concepts.md` to `2026/day-15/`

2. Commit and push to your fork

Learn in Public

Share what you learned about DNS, subnets, or ports on LinkedIn.

#90DaysOfDevOps #DevOpsKaJosh #TrainWithShubham



Happy Learning! **TrainWithShubham**